

Deterministic Finite Automaton (DFA)

machine = automaton

↳ one = automaton (tekil)

↳ two = automata (göçül)

Example! Did a heatwave occur?

Input: String of weather data

* Heatwave: temperature $\geq 45^\circ\text{C}$ for 2 consecutive days



LANGUAGE OF THE MACHINE

$L_M = \{ \text{all strings containing } 11 \}$

M accepts 11.

Also M accepts 110, 0110, 1011, 1010110.

WHY we call deterministic?

↳ Because the next steps are all determined.

Formal Definition of $M = (Q, \Sigma, \delta, q_0, F)$

is the set of states

(q_0, q_1, q_2)

is the alphabet

$\Sigma = \{1, 0\}$

is the transition function

	1	0
q_0	q_1	q_0
q_1	q_2	q_1
q_2	q_2	q_2

↳ q_0 : Start State

↳ F : set of accept/final states = $\{q_2\}$

Regular Language

A language recognized by some finite automaton

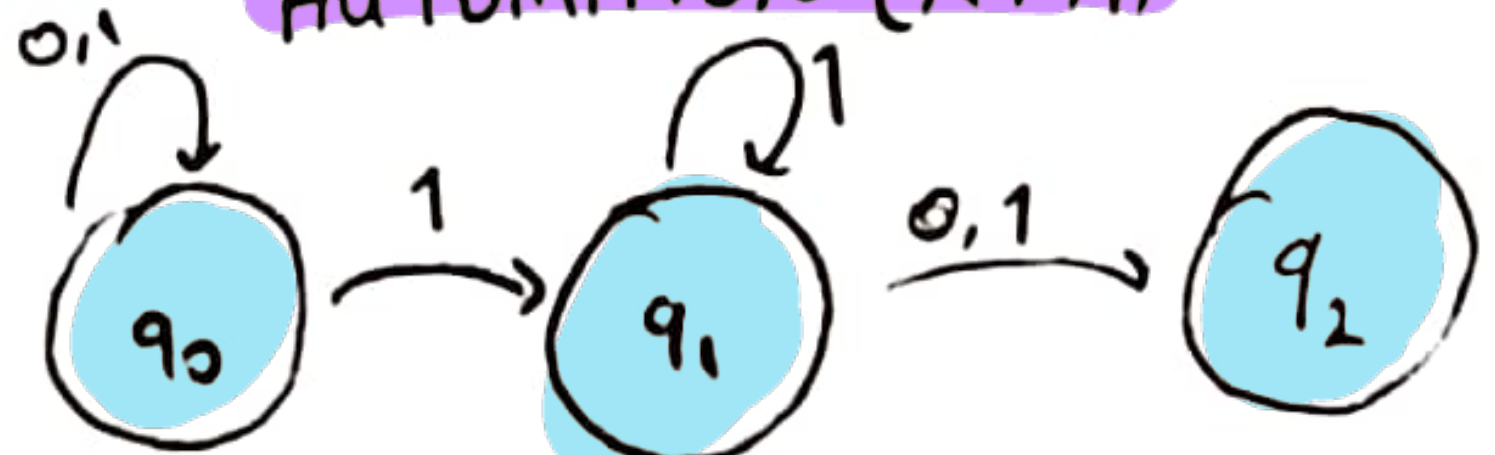
SUMMARY

DFA's are 5-tuples $(Q, \Sigma, \delta, q_0, F)$

⊙ = This means accept state

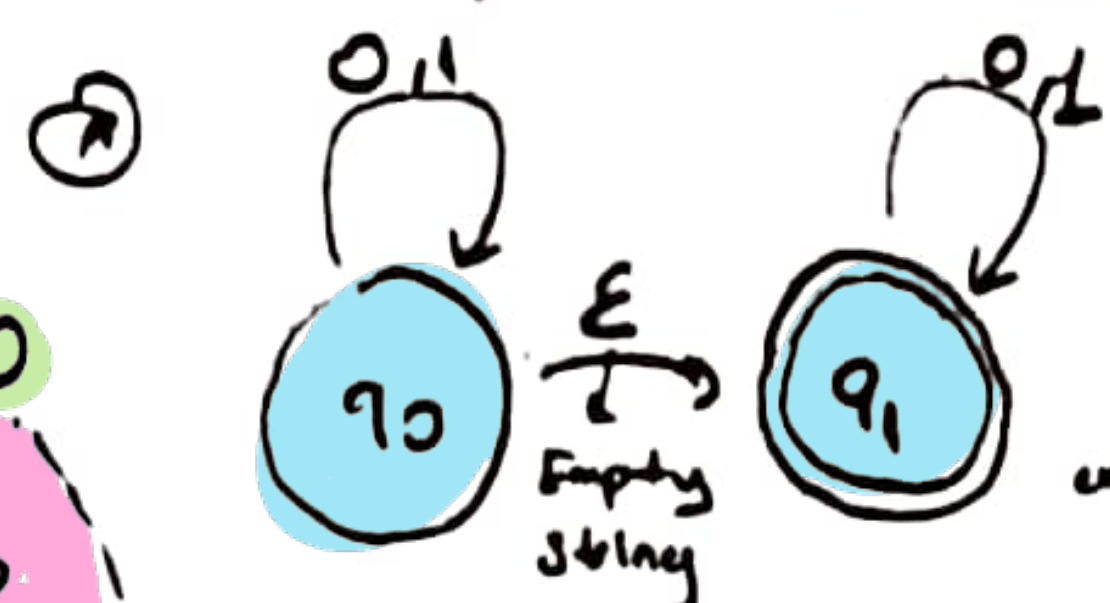
L_M is set of all accepted strings

NONDETERMINISTIC FINITE AUTOMATON (NFA)



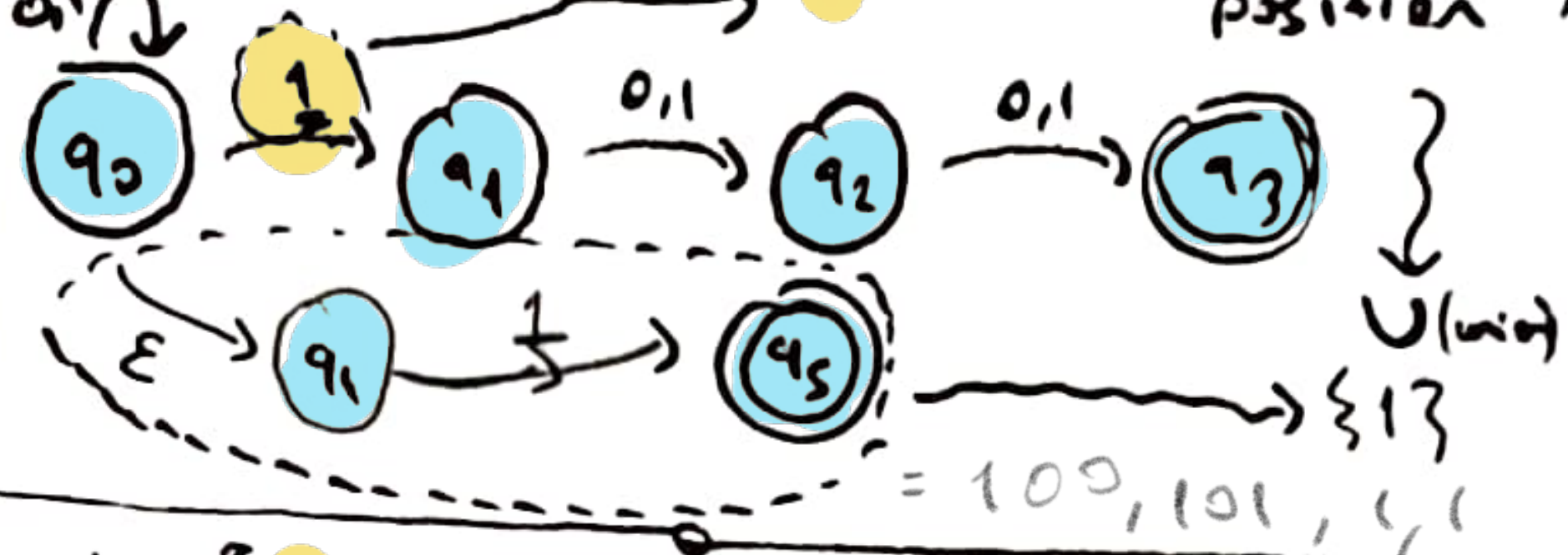
INPUT: 11 = goes two different paths.

The NFA still accepts the string, because there is a path to an accept state.



Input: 01
Notice there is an empty string in here

* $L_M = \{ x \mid x \text{ contains a } 1 \text{ in the third final position} \}$



$M = (Q, \Sigma, \delta, q_0, F)$ delta

$\{q_0, q_1, q_2, q_3, q_4, q_5\}$

$\Sigma = \{1, 0\}$

q_0 : start

$F: \{q_3, q_4\}$

$\delta: Q \times \Sigma \rightarrow P(Q)$

$\Sigma \cup \{\epsilon\}$

* δ gives a set of possible states, instead of just 1

$\delta:$	0	1	ϵ
q_0	$\{q_0\}$	$\{q_0, q_1\}$	$\{q_4\}$

NFA can only do as much as DFA can so DFA is more powerful

SUMMARY

NFA's are 5-tuples $(Q, \Sigma, \delta, q_0, F)$



That means set of possible states

• Languages recognized by an NFA regular languages